

★ Research Summary

- PhD researcher at IIIT-Delhi specialising in HCI, multimodal AI, and deep learning systems for inclusive education. First-author publications at Scientific Reports, CHI (CORE A*), AIED (CORE A), and MMM. Research addresses an underserved and clinically significant problem — building intelligent systems to support children with ADHD in digital learning environments, a population of 150M+ worldwide. Experienced in multimodal ML, computer vision, NLP, and physiological signal processing. Recipient of the Gary Marsden Award (CHI 2025) and CHI Travel Award (2025).

🎓 Education

IIIT-Delhi, Delhi

Started in Jan, 2022

PhD (Thesis: *Decoding Attention in Children with ADHD through Multimodal Analysis for Digital Learning*)

Madan Mohan Malaviya University of Technology, Gorakhpur

2016

M-Tech

Moradabad Institute of Technology, Moradabad

2013

B-Tech

🧰 Research Experience

Doctoral Researcher – Human–Machine Interaction Lab, IIIT-Delhi, India

Jan 2022 – Expected Jan 2027

- **Developed CogFormer, a task-conditioned multimodal model for sequential gaze prediction**, integrating visual-language representations through conditional fusion, self-attention, LSTM-MDN sequence modelling, and dynamic inhibition-of-return; achieved 0.735 ScanMatch vs. 0.306 for the corresponding baseline on CogGaze.
- **Investigated attention dynamics from 1000-Hz pupillometry**, modelling trial-level pupil trajectories and within-/between-trial variability in children with and without ADHD; findings on attention-related variability and medication-associated modulation. Published in Scientific Reports, 2026.
- **Developed real-time spatiotemporal attention-inference pipelines** for digital learning using webcam-derived behavioural cues and deep-learning models, targeting continuous estimation of learner attention.
- **Developed a multimodal wearable-sensing framework for ADHD characterisation** by integrating time-aligned wrist actigraphy and ECG-derived HRV over 1,021+ hours from 50 participants; multimodal fusion improved participant-independent classification, achieving 80% accuracy on the held-out test split.

Research Intern – National Institute of Informatics (NII), Japan

Oct 2023 – Feb 2024

- **Developed and evaluated multimodal vision-language methods** combining visual and textual representations for textbook-image classification during a research internship at NII, Japan; the work resulted in a peer-reviewed publication at MMM 2025.

📄 Selected Publications

• PhD-Related Research

- **S. Yadav**, A. A, J Shukla, “Trial-level pupil dynamics reflect attention-related variability and medication-associated modulation in ADHD.” *Scientific Reports*, 2026. (Q1 Journal)
- **S. Yadav**, A Upadhyay, J Shukla, “EngageME: Real-Time Spatiotemporal Attention Inference in Digital Learning Environments,” in *International Conference on Technology 4 Education (T4E)*, 2025. (Short Paper)
- **S. Yadav**, “Decoding Attention in Children with Attention Deficit Hyperactivity Disorder through Multimodal Analysis for Digital Learning,” in *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pp. 1–6, ACM, 2025. (Doctoral Consortium Paper) [CORE A*]
- **S. Yadav**, É. Lincker, C. Huron, S. Martin, C. Guinaudeau, S. Satoh, and J. Shukla, “Towards Inclusive Education: Multimodal Classification of Textbook Images for Accessibility,” in *International Conference on Multimedia Modeling (MMM)*, pp. 212–225, Springer, 2025. [CORE B]

- **S. Yadav**, M. N. Siddiqui, Y. Vats, and J. Shukla, “Engageme: Exploring Neuropsychological Tests for Assessing Attention in Online Learning,” in *International Conference on Artificial Intelligence in Education (AIED)*, pp. 394–408, Springer, 2024. [Core A]
- **S. Yadav**, É. Lincker, C. Huron, S. Martin, C. Guinaudeau, S. Satoh, and J. Shukla, “Towards Inclusive Pedagogy: A Multimodal Classification of Textbook Illustrations for Adapted Learning Environments,” in *JEP-TALN-RECITAL*, 2024. (Poster)
- **S. Yadav**, M. N. Siddiqui, and J. Shukla, “EngageMe: Assessing Student Engagement in Online Learning Using Neuropsychological Tests,” in *23rd International Conference on Artificial Intelligence in Education (AIED)*, pp. 148–154, Springer, 2023. (Doctoral Consortium Paper) [CORE A]

• Journal Publications (Pre-PhD)

- **S. Yadav**, N. Sengar, A. Singh, A. Singh, and M. K. Dutta. Identification of disease using deep learning and evaluation of bacteriosis in peach leaf. *Ecological Informatics*, 61: 101247. 2021. [Q1, IF = 5.9]
- Joshi, R.C., **Yadav S.**, Pathak, V.K., Malhotra, H.S., Khokhar, H.V.S., Parihar, A., Kohli, N., Himanshu, D., Garg, R.K., Bhatt, M.L.B. and Kumar, R. A deep learning-based COVID-19 automatic diagnostic framework using chest X-ray images. *Biocybernetics and Biomedical Engineering*, 41(1), pp.239-254. 2021 [Q1, IF = 5.3]

• Book Chapters

- **Yadav S.**, Chandra Joshi D., Joshi A., Mathur S., and Bhatt M.. "Intelligent approaches for disease detection and prevention." In *Internet of Things in Modern Computing*, pp. 113-132. CRC Press, 2023.
- **Yadav S.**, Joshi, R. C., Yadav, D. Trustworthy machine learning for cloud-based internet of things (IoT). In *Transforming Management with AI, Big-Data, and IoT* (pp. 155-167). Cham: Springer International Publishing. 2022

Work Experience

- Teaching Fellow, IIIT-Delhi | January 2021–December, 2021.
- Junior Research Fellow (JRF), Centre for Advanced Studies, AKTU | July 2019–December 2020.
- Assistant Professor, IEC College of Engineering and Technology | January 2018–June 2019.

Grant and Awards

- Gary Marsden Award, CHI 2025 — Recognises outstanding HCI researchers advancing computing for underserved communities.
- CHI Travel Award, 2025 — Competitive award for first-author researchers accepted to ACM CHI (CORE A*).
- AIED Conference Scholarship, 2024 & 2023 — Merit-based award for distinguished doctoral contributions to AI in Education.
- Selected: ACM India Women in Computing Cohort, 2025 & 2026 — Competitive cohort recognising women researchers making an impact in computing.
- Selected: IndoML Symposium, Poster Presentation + Travel Grant, 2024 and 2026 — National ML symposium; awarded travel grant for accepted poster.
- Qualified NTA UGC NET, 2019 — National-level eligibility exam for research and teaching positions in India.

Technical Skills

- **Programming Languages:** Python, C.
- **ML& DL:** Scikit-learn, TensorFlow, Keras, PyTorch, CNNs, Transformers.
- **Computer Vision:** OpenCV, YOLO, ViT; image classification, object detection, multimodal fusion

Saumya Yadav

☎ +91-7905814587

🏠 Delhi, India

🌐 saumyadav.github.io

✉ saumya@iiitd.ac.in

in [/saumya-yadav-152021](https://www.linkedin.com/in/saumya-yadav-152021)

🎓 [Google Scholar](#)

- **NLP** HuggingFace Transformers (BERT, CamemBERT, GPT-based); text classification, cross-lingual modelling, prompt engineering
- **Signal Processing & Biosensors:** EEG preprocessing, HRV analysis, eye-tracking data pipelines
- **Explainability:** SHAP, LIME
- **Tools & Platforms:** Git, Jupyter, Google Colab, LaTeX

I hereby declare that the above information is true to my knowledge.

SAUMYA YADAV